Informal learning in everyday human–LLM interaction

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Abstract

As LLM-based assistants become everyday companions for work and study, lifelong learning is increasingly mediated by everyday human–LLM interaction. Yet it remains unclear whether these interactions elicit the engagement through which people build understanding. We analyse 128,334 naturalistic conversations from three public conversational corpora—WildChat, LMSYS Chat-1M and ShareChat strict-English—adapting learning-science constructs to 979,974 turns through human-verified LLM-assisted annotation. Across 490,932 user turns, cognitive engagement appeared in 31.9% and constructive engagement—the stricter level in which users elaborate, test, revise or extend ideas—in 4.9%. Constructive engagement was shaped by learning framing, task ecology and sustained exchange across the three corpora. On the other side, assistant support was most informative not as a binary scaffolding signal but as a situated match between support form and user activity: feedback-like and explanatory forms aligned with constructive engagement, and adjacent-turn analyses showed local, state-dependent coupling rather than automatic accumulation. These findings make informal learning in everyday AI use measurable, laying a foundation for personal assistants that help people build knowledge while solving real problems.

Keywords: human–LLM interaction, informal learning, large language models, scaffolding, learner engagement

1 Introduction

Large language models (LLMs) are becoming embedded in the cognitive work of everyday life. People use them to write, code, search, troubleshoot and make sense of unfamiliar problems, and field and experimental evidence increasingly treats generative AI as part of ordinary knowledge work rather than only as a classroom technology [1–3]. Most of these interactions do not take place in classrooms, training programmes or purpose-built tutoring systems. Instead, they arise in the course of routine activity, when users turn to LLMs to solve an immediate problem, clarify a concept, improve a draft or navigate an unfamiliar task. In such moments, the model may do more than help complete the task at hand: it may shape what the user notices, understands and tries next. Everyday human–LLM interaction therefore constitutes a rapidly expanding setting for informal learning—that is, learning that occurs outside formal curricula and designed instructional environments, and instead emerges through everyday activity, self-directed inquiry and problem solving [4]. This learning may be intentional, when users explicitly seek understanding, or incidental, when understanding develops while pursuing another immediate goal [5, 6].

Yet existing research on LLMs and learning has focused primarily on settings in which learning support is deliberately designed, such as automated feedback, question generation, retrieval practice and tutoring systems [7–11]. These studies show that LLMs can enact instructional strategies when they are explicitly prompted, trained or embedded in purpose-built systems [12, 13]; they tell us less about whether learning-oriented engagement emerges when general-purpose LLMs are used in the course of everyday tasks.

This gap matters because the educational significance of everyday AI use depends not only on what models can provide, but on what users do with that support. The key question is not simply whether LLMs can generate explanations, examples or feedback, but whether these interactions keep users active in the work of understanding: asking questions, testing assumptions, revising ideas and extending solutions. Answering this question would open a path toward future personal AI assistants that function not only as tools for completing tasks, but also as context-sensitive companions for lifelong learning. Such systems would need to recognize when a user is ready for feedback, explanation, modelling or direct help, and when support might instead short-circuit the user's own sense-making. Conversely, without evidence about these interactional dynamics, it is difficult to know whether AI-mediated work is helping people build knowledge and skill over time, or merely making answer delivery faster while leaving human understanding opaque. We therefore ask how often users show learning-oriented engagement in everyday human–LLM interaction, and which contextual and interactional conditions make it more likely.

Studying this question in naturalistic logs requires adapting educational theory to data without fixed instructional tasks, explicit learning objectives or post-test outcomes. We translate theories of engagement and scaffolding into role-specific, turn-level behavioural signatures: user turns are coded for passive, active or constructive uptake [14, 15], while assistant turns are coded for the presence of scaffolded support and further characterized by support intent and support form [16–19]. Following the

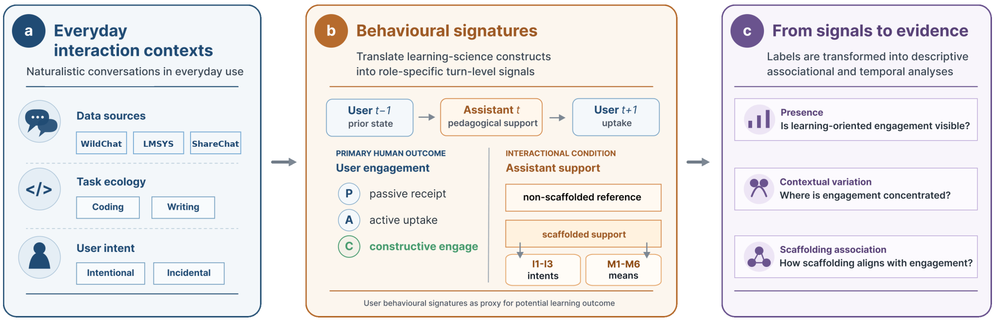


Fig. 1 Analytic framework for studying informal learning in everyday human–LLM interaction. a, Naturalistic public conversations are situated by task ecology and user framing. **b,** Learning-science constructs are translated into turn-level behavioural signatures for user engagement and assistant support. **c,** The resulting labels support analyses of engagement presence, contextual variation and scaffolding–engagement association. Labels capture observable interactional signatures, not durable learning outcomes or causal instructional effects.

literature that deeper forms of overt engagement are more strongly linked to learning [15], we treat constructive engagement as the strictest observable indicator of learning-oriented participation available in naturalistic logs. These measures do not establish causal learning effects or durable knowledge gains; instead, they allow us to observe whether learning-oriented engagement is present in everyday use and to test which contextual and interactional conditions are associated with deeper user participation.

Using this adapted framework, we analyse 128,334 naturalistic conversations from three public conversational corpora: WildChat, LMSYS Chat-1M and ShareChat strict-English [20–22]. Across these corpora, we focus on coding and writing as two common task ecologies of everyday LLM use. The corpus contains 490,932 user turns and 489,042 assistant turns. The analysis proceeds in three layers: first, whether learning-oriented engagement is visible and how it is composed; second, where constructive engagement is concentrated across user framing, task ecology and interaction depth; and third, how assistant scaffolding is associated with constructive participation across conversation-level, support-form and adjacent-turn temporal scales.

We find, first, that learning-oriented engagement is a recurrent feature of everyday human–LLM interaction, although its constructive form is selective. Across the three public corpora, cognitive engagement appears in 31.9% of user turns and constructive engagement in 4.9%, while emotional engagement is rare; scaffolded support appears in 31.7% of assistant turns. These rates show that everyday AI use often involves more than answer delivery, but the most generative and constructive form of user engagement is a stricter and less frequent behavioural signal. Second, constructive engagement is organized by context: it is amplified by explicit learning framing, varies across task ecologies and becomes most visible in sustained exchanges where users return to, interrogate and reshape the assistant's response. Third, assistant scaffolding marks richer constructive participation across the reportable public-chat settings, but not as a uniform or treatment-like effect. Its association with engagement depends on user framing, support form, task-specific support supply and temporal scale: feedback-like and explanatory support align most clearly with constructive engagement, whereas more directive or questioning forms are less aligned with visible elaboration, and the strongest temporal evidence is local and state-dependent rather than cumulative. Together, these findings locate everyday human–LLM interaction in a middle space between ordinary tool use and designed instruction. By making this middle space observable, the study provides a foundation for future personal AI assistants that support lifelong learning in the flow of everyday work—not by turning every interaction into a formal lesson, but by helping everyday human–AI collaboration become a site where people build, test and extend understanding while solving real problems.

2 Results

The results proceed from engagement presence to the conditions that differentiate its depth. We first ask whether everyday human–LLM conversations contain learning-oriented engagement, distinguishing broad cognitive uptake from the stricter constructive form. We then examine where constructive engagement is concentrated across user framing, task ecology and sustained exchange. Finally, we ask how assistant scaffolding relates to constructive engagement: first as a conversation-level condition, then through support forms, and finally through adjacent-turn temporal ordering. Because the logs contain no direct assessment of knowledge gain, engagement labels are interpreted as behavioural signatures of potential learning, and scaffolding analyses are interpreted as associations rather than causal evidence of learning effects.

2.1 Everyday human–LLM conversations contain learning-oriented engagement

Across the six WildChat, LMSYS Chat-1M and ShareChat strict-English task settings, 31.9% of user turns showed cognitive engagement, 4.9% reached constructive engagement, and emotional engagement was rare, appearing in only 0.74% of user turns (Table 1). This establishes the first descriptive pattern: users often engage with assistant responses in learning-oriented ways rather than merely passively acknowledging or receiving task output, while their affective engagement is limited.

Figure 2a decomposes cognitively engaged turns into passive receipt, active uptake and constructive engagement across the six public-chat task settings. Active uptake was the dominant form of cognitive engagement, while the stricter constructive level formed a smaller but stable subset. A majority of cognitively engaged turns involved active uptake: users used, applied or followed up on the assistant's response. A smaller but stable subset involved constructive engagement, in which users elaborated, tested, revised or extended ideas. This distinction matters for the rest of the Results. Cognitive engagement provides the broad signal that everyday LLM use can become learning-oriented, active uptake shows that users often work with the assistant's response, and constructive engagement identifies the stricter cases of deeper user participation that we seek to explain. The following analyses therefore focus on where this constructive form appears and which interactional conditions are associated with it.

Table 1 Dataset and behavioural overview. Summary statistics for the WildChat, LMSYS Chat-1M and ShareChat strict-English task settings analysed in the study. Constructive engagement is reported as the stricter depth level within cognitive engagement and serves as the focal user-side outcome in subsequent analyses.

	Sample size			User framing	Assistant support	Cognitive engagement		Emotional engagement
Setting	Conversations	User turns	Assistant turns	Intentional conv. (%)	Scaffolded turns (%)	Overall (%)	Constructive level (%)	Emotional (%)
WildChat coding	31,878	124,073	124,623	37.13	51.03	50.21	9.23	0.95
WildChat writing	39,534	163,705	164,824	25.80	23.53	16.34	2.22	0.57
LMSYS coding	31,879	106,312	104,431	41.54	29.11	45.28	5.43	0.90
LMSYS writing	21,023	77,906	76,279	20.64	19.43	15.30	1.53	0.21
ShareChat coding	2,481	11,541	11,522	45.63	50.67	45.00	15.17	3.38
ShareChat writing	1,539	7,395	7,363	29.69	22.42	33.05	5.18	0.34
Total / pooled	128,334	490,932	489,042	32.11	31.71	31.93	4.93	0.74

Note. Percentages for cognitive, constructive and emotional engagement are computed over user turns; percentages for scaffolded support are computed over assistant turns. The constructive-level column reports the share of all user turns that reached the constructive depth level within cognitive engagement. Figure 2a reports the composition of passive, active and constructive uptake among cognitively engaged turns.

2.2 Constructive engagement is shaped by user framing, task ecology and interaction depth

Having established that everyday human–LLM conversations contain learning-oriented engagement, we next examined where its constructive form was concentrated. User framing provided the first contrast (Fig. 2b). Across the six main public-chat settings, conversations explicitly oriented toward learning or exploration showed higher engagement than conversations without such framing. Importantly, both explicitly learning-oriented and more task-oriented conversations contained constructive participation, indicating that informal learning-relevant engagement was not confined to interactions that announced themselves as learning episodes.

Task ecology provided a second source of variation. In the two everyday task ecologies sampled here, coding-oriented conversations showed higher constructive engagement than writing-oriented conversations across WildChat, LMSYS and ShareChat. Cognitive engagement was 45.00–50.21% in coding conversations, compared with 15.30–33.05% in writing conversations; constructive engagement was 5.43–15.17% in coding and 1.53–5.18% in writing. The LMSYS coding setting illustrates why broad cognitive uptake and deeper constructive engagement should remain separate outcomes: because LMSYS-Chat-1M reflects an earlier public-demo and Chatbot Arena period [21], many coding exchanges involved active repair questions, added constraints and requests for another direct coding step, which can raise active uptake without producing the same level of constructive elaboration. One plausible reason for a higher engagement rate of coding tasks is that they often make uncertainty and failure externally visible through errors, constraints and executable tests, whereas writing tasks may more often be resolved through drafting, editing or applying a proposed revision without requiring users to articulate the underlying rationale [23–25]. This pattern suggests that task nature shaped the baseline opportunities for users to ask questions, test assumptions, refine solutions and extend ideas, therefore influencing users' engagement levels.

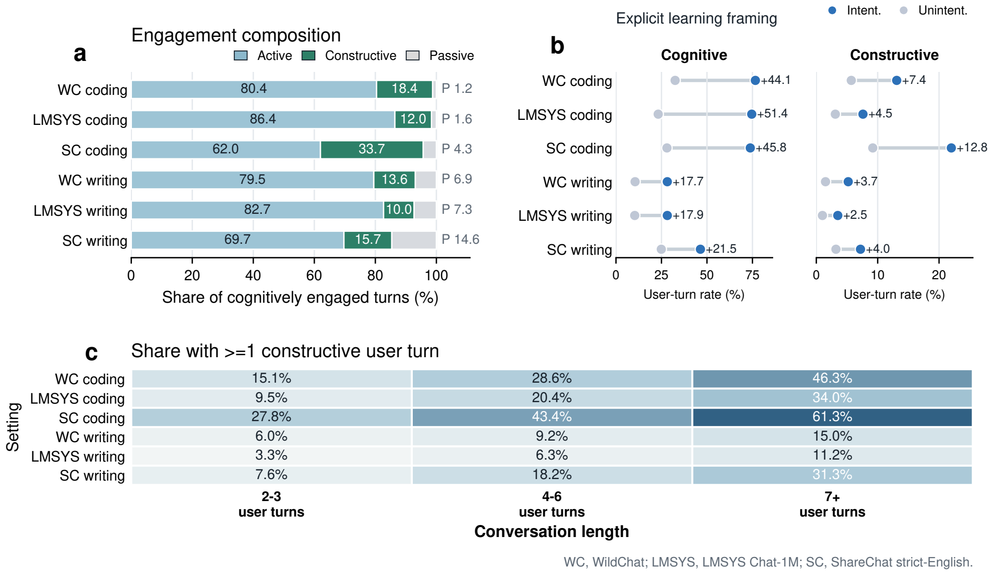


Fig. 2 Learning-oriented engagement varies by engagement form, user framing and conversation length. a, Composition of cognitively engaged user turns by active, constructive and passive levels across the six public-chat task settings. **b,** Cognitive and constructive engagement rates in conversations with versus without explicit learning framing. **c,** Share of conversations with at least one constructive user turn by conversation-length bucket. Panel a uses cognitively engaged turns as the denominator; panel b reports percentages of user turns; panel c reports conversation-level shares. WC, WildChat; LMSYS, LMSYS Chat-1M; SC, ShareChat strict-English.

Interaction depth provided a third lens on where constructive participation surfaced (Fig. 2c). Across the six main public-chat settings, the probability that a conversation contained at least one constructive user turn increased with behavioural depth. The gradient was steeper in coding-oriented conversations, but the same direction appeared in writing-oriented conversations as well. This pattern shows that constructive engagement is most visible when everyday LLM use becomes a sustained exchange rather than a one-off request for an answer. In such exchanges, users have room to return to the assistant's response, introduce new constraints, test implications, request revisions and reshape their understanding.

Together, these results show that constructive engagement is ecological. It is amplified by explicit learning framing, varies across task ecologies and concentrates in sustained exchanges.

2.3 Scaffolded conversations show richer constructive participation

After characterizing user-side conditions such as learning intent and task ecology, we next examined how assistant behaviour was associated with richer constructive participation through scaffolding, a canonical way that teachers support engagement. At the conversation level, conversations containing at least one scaffolded assistant

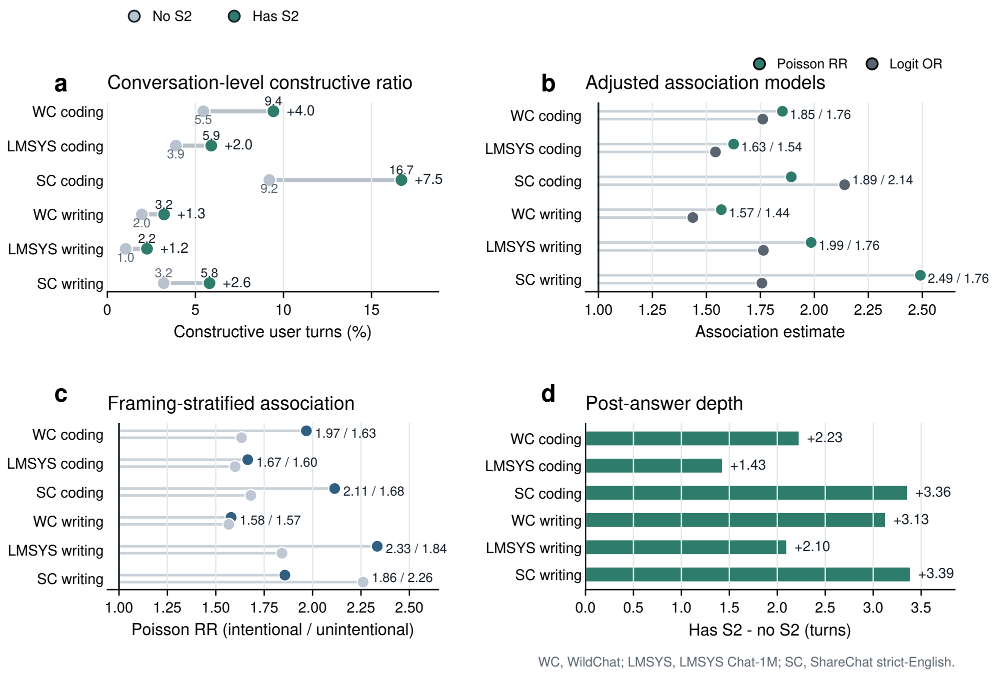


Fig. 3 **Scaffolded conversations are associated with richer constructive participation across public-chat corpora. a,** Constructive user-turn ratios in conversations with versus without scaffolded support. **b,** Covariate-adjusted associations between scaffolded-support presence and constructive participation, reported as Poisson rate ratios and logistic odds ratios. **c,** Poisson rate ratios for scaffolded-support presence stratified by intentional versus unintentional user framing. **d,** Difference in post-answer interaction depth after the first assistant response. Numbers indicate scaffolded-minus-reference differences, except panels b and c, which report model estimates. Lines in panels b and c are visual guides from the reference value to the point estimate.

turn had higher constructive ratios than conversations without scaffolded support in all six task settings (Fig. 3a). Bootstrap estimates of these raw differences ranged from +0.99 to +7.34 percentage points across WildChat, LMSYS and ShareChat, and scaffolded conversations also showed greater post-first-response depth in every setting (+1.43 to +3.38 turns; Fig. 3d). All six constructive-ratio contrasts and all six depth contrasts were statistically distinguishable from zero in conversation-level bootstrap tests ($p < .001$; Supplementary Table C6). These descriptive contrasts show that scaffolding co-occurred with both more constructive participation and more sustained exchange; the percentage-point differences are reported as effect sizes.

The association remained positive after accounting for measured user status and interactional context, including user framing, task ecology and interaction depth, with covariate-adjusted models. Across the six settings, Poisson rate ratios for constructive-turn counts ranged from 1.569 to 2.491, and logistic odds ratios for whether a conversation contained at least one constructive turn ranged from 1.437 to 2.140 (Fig. 3b). These estimates indicate that scaffolded support remained associated with higher constructive participation after measured adjustment.

Table 2 Summary of key support–engagement associations across public-chat corpora. Constructive-ratio contrasts compare conversations with versus without scaffolded support. Adjusted models report the association of scaffolded-support presence with constructive-turn count (Poisson rate ratio) and with having at least one constructive turn (logit odds ratio). Adjacent-turn lift is the difference in the probability that the next user turn is constructive after scaffolded versus non-scaffolded reference assistant turns.

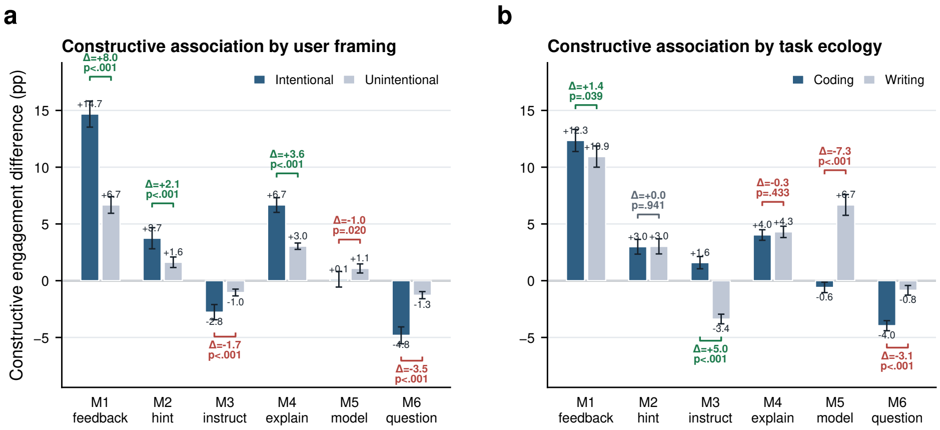
Setting	Constructive ratio Has S2 / No S2	Difference (pp)	Depth diff. (turns)	Poisson RR	Logit OR	Adjacent lift (pp)	Around first S2 diff.
WildChat coding	0.094 / 0.055	+3.9	+2.23	1.852	1.761	+2.68	+0.014
LMSYS coding	0.059 / 0.039	+2.0	+1.43	1.626	1.542	+2.01	+0.015
ShareChat coding	0.167 / 0.092	+7.5	+3.36	1.893	2.140	+6.21	+0.007
WildChat writing	0.032 / 0.020	+1.2	+3.13	1.569	1.437	+1.13	-0.042
LMSYS writing	0.022 / 0.010	+1.2	+2.10	1.985	1.764	+0.27	-0.025
ShareChat writing	0.058 / 0.032	+2.6	+3.39	2.491	1.757	+3.35	-0.006

2.4 Support forms differentiate constructive engagement

Education theory treats scaffolding as a family of support moves rather than a single uniform intervention. We next examined how different forms of scaffolded support were associated with users' constructive engagement. We decomposed scaffolded support into the six support forms in the annotation scheme (M1–M6) and, among scaffolded conversations, compared conversations containing each form with scaffolded conversations without that form (Fig. 4a). Because support-form labels were non-mutually exclusive, these estimates describe form-level associations rather than effects of isolated instructional moves.

The form-level pattern was sharper than the binary distinction between scaffolded and non-scaffolded support. When stratified by user framing, feedback-like support showed the largest constructive-engagement contrast in both intentional and unintentional conversations, with a stronger association under intentional framing (+14.7 versus +6.7 percentage points; framing difference, +8.0 points; Fig. 4a). Explanation showed the same direction (+6.7 versus +3.0 points; framing difference, +3.6 points), whereas instructing and questioning were lower, especially in intentionally framed conversations. Task-stratified estimates provided a complementary check (Fig. 4b): feedback and explanation were positive in both coding- and writing-oriented ecologies, while instructing, modelling and questioning varied more strongly by task. This pattern suggests that the forms most aligned with constructive participation were those that responded to a user's prior attempt or made mechanisms available for inspection, while more directive, example-based or clarification-oriented moves depended more strongly on the surrounding task ecology. Thus, after the conversation-level association in Section 2.3, the finer drill-down is not that all scaffolding is equally useful, but that support form shapes whether scaffolding leaves a productive role for the user in the work of understanding.

Assistant support-intent labels added little separation because most scaffolded support was cognitively oriented. The more informative distinction was therefore how support was delivered. Figure 4c situates this form-level pattern within task ecology: explanation dominated scaffolded coding turns across the three corpora, whereas writing-oriented support supply was more mixed and more often included instructing, explanation or questioning forms.



c Supply within scaffolded assistant turns (%)

	M1	M2	M3	M4	M5	M6
WC coding	8.8	14.7	29.7	84.0	23.9	6.9
LMSYS coding	4.5	8.2	10.4	79.2	18.3	10.6
SC coding	13.3	12.7	25.7	81.2	18.4	18.6
WC writing	16.6	9.6	59.2	29.5	15.0	6.9
LMSYS writing	11.3	10.9	52.3	26.9	10.3	13.4
SC writing	24.2	23.2	22.9	45.2	23.5	38.1

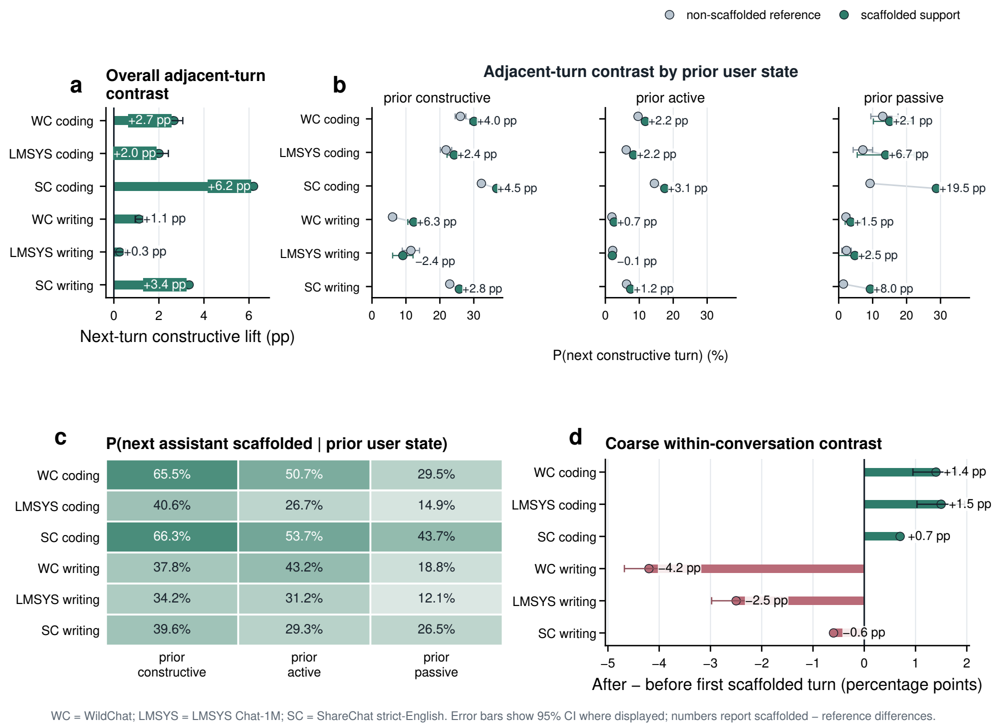
WC, WildChat; LMSYS, LMSYS Chat-1M; SC, ShareChat strict-English. Means labels are non-exclusive.

Fig. 4 Support form and support supply differentiate scaffolded interaction. a, Constructive-engagement differences associated with each support form among scaffolded conversations, stratified by intentional versus unintentional user framing. **b,** The same support-form contrasts stratified by coding- versus writing-oriented task ecology. **c,** Percentage of scaffolded assistant turns containing each support form by task setting. In panels a and b, bars compare scaffolded conversations containing a support form with scaffolded conversations without that form; error bars show 95% confidence intervals, and bracket annotations show between-stratum differences and two-sided p values. Support-form labels correspond to M1–M6 and are not mutually exclusive, so rows do not sum to 100%.

Together, these results show why scaffolding presence alone is too coarse to characterize learning-oriented human–LLM interaction. Scaffolded support was associated with richer constructive participation at the conversation level, but that association was carried unevenly by support form and task-specific support supply. The implication is not that systems should simply provide more scaffolding: feedback and explanation may sustain users who are already working with ideas, whereas more directive forms may complete the task without eliciting further elaboration.

2.5 Support–engagement coupling is local and state-dependent

Conversation-level models showed that scaffolded conversations contained richer constructive participation. The temporal question is different: how is this support–engagement association expressed over interaction time? It could appear as local coupling, in which scaffolded support is followed by constructive engagement in the next user turn; or it could appear as a cumulative trajectory, in which constructive



	prior constructive	prior active	prior passive
WC coding	65.5%	50.7%	29.5%
LMSYS coding	40.6%	26.7%	14.9%
SC coding	66.3%	53.7%	43.7%
WC writing	37.8%	43.2%	18.8%
LMSYS writing	34.2%	31.2%	12.1%
SC writing	39.6%	29.3%	26.5%

Fig. 5 Support–engagement coupling is local and state-dependent across public-chat corpora. a, Next-turn constructive lift after scaffolded versus non-scaffolded reference assistant turns. **b,** Probability of a constructive next user turn by prior user state and support condition; annotations show scaffolded-minus-reference differences. **c,** Probability that the next assistant turn contains scaffolded support by prior user state. **d,** Constructive-engagement contrast after versus before the first scaffolded assistant turn. Error bars show 95% confidence intervals where displayed. WC, WildChat; LMSYS, LMSYS Chat-1M; SC, ShareChat strict-English.

engagement rises after scaffolding first appears in a conversation. Figure 6 summarizes these two temporal estimands.

At the adjacent-turn scale, scaffolded assistant turns were more likely than non-scaffolded reference turns to be followed by constructive engagement in all six task settings (Fig. 5a). The lift was positive in each setting, ranging from +0.27 to +6.21 percentage points. Five of the six adjacent-turn lifts were statistically distinguishable from zero in conversation-cluster bootstrap tests; LMSYS writing was positive but weaker and did not cross the conventional .05 threshold (+0.27 percentage points, $p =$ .061; Supplementary Table C6). This result gives the most direct temporal evidence for support–engagement coupling: the association appeared in the immediately following user turn, although its magnitude varied across public-chat settings.

The local signal was state-dependent. State-conditioned contrasts were generally positive but varied in magnitude depending on the immediately preceding user state (Fig. 5b). The reverse ordering showed the same state sensitivity: assistant scaffolding was more likely after constructive or active user turns than after passive turns,

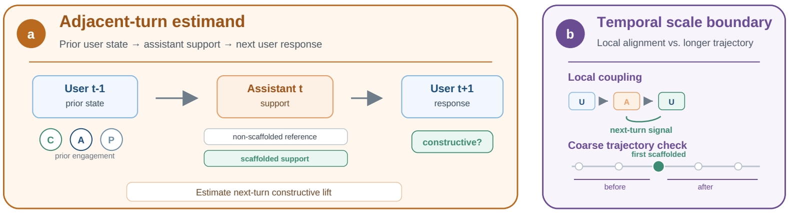


Fig. 6 **Temporal estimands for support–engagement coupling. a,** Adjacent-turn analysis represents each local sequence as a prior user state, an assistant support condition and the next user response. **b,** Coarser within-conversation analysis compares constructive engagement after versus before the first scaffolded assistant turn. The schematic separates local support–engagement coupling from broader accumulation over a conversation.

especially in coding-oriented conversations (Fig. 5c). Thus, support and engagement were coupled around local interactional states: scaffolded support often sustained an active or constructive thread, and users' current engagement shaped the support they received next.

To place user state, assistant scaffolding and adjacent-turn context in a single model, we fitted an integrated adjacent-turn logistic regression predicting whether the next user turn was constructive. In the pooled six-setting model with dataset fixed effects, prior user state was the strongest predictor (prior constructive, odds ratio (OR) 8.24, 95% CI 7.81–8.69; prior active, OR 2.47, 95% CI 2.37–2.57), and explanatory support (M4) also remained strongly positive (OR 2.15, 95% CI 1.99–2.32). By contrast, the broad scaffolded-support indicator was not statistically distinguishable from the null once prior state and specific support forms were included (OR 0.94, 95% CI 0.87–1.02, $p = .164$). A sensitivity replacing dataset fixed effects with recoverable model/source fixed effects, and a WildChat-only sensitivity with exact model fixed effects, preserved the same substantive pattern while showing that model/source labels add detectable heterogeneity (Supplementary Table C7). This joint model therefore supports a narrower interpretation: local state and the form of scaffolding, rather than scaffolding presence alone or model identity alone, best summarize the adjacent-turn coupling signal.

The longer-window contrast located the boundary of this pattern. Comparing constructive engagement after versus before the first scaffolded assistant turn yielded small positive contrasts in coding and negative contrasts in writing (Fig. 5d). This coarser contrast did not show the same pattern as the adjacent-turn analysis. The support–engagement signal was therefore strongest at the local, state-conditioned scale: everyday LLM interaction creates repeated moments in which support and constructive engagement align, but those moments do not automatically form a single conversation-level trajectory.

Together, the temporal analyses sharpen the boundary of the support result. Scaffolded support was associated with constructive participation, but the strongest temporal evidence was local, state-dependent and task-ecology specific. Everyday human–LLM interaction therefore forms a coupled interaction process: users' current engagement shapes the support they receive, and scaffolded support is locally associated with subsequent constructive engagement. Learning-oriented engagement in everyday LLM use therefore depends on situated sequences in which task ecology, user framing, support form and user state jointly shape whether people continue working constructively with ideas.

3 Discussion

Everyday human–LLM interaction as an informal learning ecology

This study identifies everyday human–LLM interaction as a middle space between ordinary tool use and designed instruction. Across naturalistic WildChat, LMSYS Chat-1M and ShareChat strict-English coding and writing conversations, learning-oriented engagement was recurrent, but its constructive form was selective. Users often used, applied or followed up on assistant responses; deeper constructive engagement emerged when the exchange preserved reasons and opportunities for users to keep working with ideas—through explicit learning framing, task affordances that made testing and revision salient, and support that left room for user evaluation rather than simply completing the task. The assistant-side results add a complementary picture: scaffolded support was associated with richer constructive participation when its form, timing and surrounding user state supported continued sense-making. Feedback and explanation aligned more clearly with constructive engagement than more directive or question-based moves, and temporal evidence was strongest in adjacent-turn coupling. Together, these findings suggest that informal learning in everyday LLM use emerges through situated exchanges in which users and assistants maintain a shared space for testing, revising and extending understanding while pursuing practical goals.

Constructive engagement in everyday AI use is ecological. User framing, task affordances and the unfolding exchange jointly configure opportunities for sense-making. Explicit learning framing amplified constructive engagement, but constructive participation also appeared in task-oriented conversations where learning was incidental. This matters because informal learning rarely announces itself as a formal learning episode. It often appears when a practical task exposes uncertainty, invites comparison or requires revision. Coding and writing illustrate this point as task ecologies rather than as isolated topics. Coding-oriented interactions often make uncertainty and failure visible through errors, constraints and executable tests; writing-oriented interactions more often involve drafting, editing or applying revisions, where the rationale for a change can remain implicit. These differences help explain why task ecology shaped the baseline opportunity for users to ask, test, refine and extend ideas. Sustained exchanges added another layer: they gave users room to return to an assistant response, introduce new constraints, test implications and reshape their understanding. In this sense, interaction depth is not merely a count of turns; it marks the conversational space in which constructive participation can surface.

Scaffolding as situated support

The support results show how assistant behaviour becomes part of this ecology. Conversations containing scaffolded support showed richer constructive participation and greater post-answer depth, and adjusted models preserved positive associations after accounting for measured conversation features. A coding-specific SWE-chat check showed the same descriptive direction in a newer software-engineering corpus, supporting the interpretation that this pattern is not limited to the three main public-chat corpora. These population-level patterns indicate that scaffolded support marks conversational environments in which constructive participation becomes more visible. The key point is not the mere presence of pedagogical language, but how support is positioned within the user's ongoing work, consistent with scaffolding accounts that emphasize contingency, responsiveness and the transfer of responsibility [16, 17, 19]. Scaffolding is most informative when analysed together with what users are trying to do, what the task makes visible and how the assistant response shapes the user's opportunity to continue working with ideas.

The most actionable insight is that support form matters. Feedback-like support and explanation were most clearly aligned with constructive engagement, whereas instructing and questioning were less aligned with visible elaboration. This distinction suggests that different support forms leave different roles for the user. Feedback responds to a user's prior attempt and can create an occasion for revision, evaluation or extension. Explanation can make mechanisms visible and give users material for testing their understanding. Instruction and questioning remain important forms of assistance, especially when users need direct task support, missing information or clarification. In the observed conversations, however, these forms were less aligned with visible constructive elaboration, suggesting that support quality for informal learning should be judged by whether it preserves a productive role for the user in the work of understanding, not only by whether it appears pedagogical in form.

This perspective also clarifies why constructive engagement is a useful organizing outcome for everyday human–LLM interaction. In ordinary use, the same system may act as an assistant, explainer, feedback provider or directive instructor within a single conversation. The learning relevance of these roles depends on whether they recruit further user thinking while helping the user make progress on the task. Constructive engagement captures the moments when users are not only receiving information but actively working with it: testing a boundary, revising an idea, articulating a constraint or extending an explanation. It therefore provides a behavioural window into the user-side work that makes informal learning visible in naturalistic conversations.

From local coupling to lifelong learning support

The temporal results locate AI-mediated informal learning at a different scale from formal instruction. Everyday LLM interactions repeatedly surface local moments in which support and user engagement align: scaffolded support is more often followed by constructive engagement, and the user's current state shapes the support that appears next. These moments are valuable because they occur at the point of need, when users are confused, blocked, comparing alternatives, repairing an attempt or adapting knowledge to a concrete problem. Formal instruction and tutoring systems contribute what local interaction alone rarely supplies: sequencing, shared standards, assessment and repeated opportunities to abstract beyond the immediate task [13]. The role of everyday AI use is therefore complementary. It can extend learning into the spaces between formal learning episodes, while future assistants can help connect repeated local episodes into longer arcs of reflection, consolidation and skill development. In this sense, the temporal results identify both the promise and the design challenge of lifelong AI assistance: preserving the immediacy and personalization of point-of-need support while giving local learning moments the continuity needed to support durable understanding, transferable skill and cumulative expertise.

This complementarity suggests a design principle for personal AI assistants that support lifelong learning: they should manage the cognitive responsibility left to the user while providing continuity across time. At the local scale, support should be calibrated to the user's state and the task affordance. Direct help may be appropriate when completion is the immediate goal, whereas explanation, feedback or partial cues are more useful when the user is trying to diagnose, evaluate or revise an idea. Equally important is restraint, because too much instruction can replace the user's own sense-making rather than support it. At the longitudinal scale, the assistant's role is to make local learning moments less ephemeral. Rather than treating each exchange as isolated, future systems could remember recurring uncertainties, link current problems to prior explanations and prompt consolidation after difficult tasks. In this sense, the distinctive promise of lifelong AI assistance is not to convert everyday work into formal instruction, but to give continuity, memory and personalization to the informal learning that happens between formal learning episodes. Together with formal instruction, such systems could support a more continuous and personalized learning ecosystem in which curricula provide structure, standards and shared progression, while personal AI assistants carry learning into everyday practice, helping people transform formal knowledge into situated judgment, transferable skill and cumulative understanding.

Scope and future directions

Several boundaries define the scope of these claims. The analyses characterize population-level patterns in three public conversational corpora, WildChat, LMSYS Chat-1M and ShareChat strict-English. They make visible where learning-oriented engagement appears, how constructive participation is organized by task ecology and user framing, and how assistant support aligns with engagement across conversation-level, support-form and adjacent-turn timescales. These patterns are useful for understanding the interactional conditions under which constructive engagement becomes more visible in everyday use, but they should be read as observational digital-trace evidence rather than causal estimates [26, 27]. Applying them to individual learners will require additional information about prior knowledge, goals, motivation and interaction history, because the same support form may function differently for different users over time. The labels also capture behavioural manifestations of learning-oriented engagement rather than direct evidence of durable learning outcomes. Constructive engagement marks observable elaboration, testing, revision or extension; future work should connect these behavioural signatures to independent measures of retention, transfer, skill development and learner autonomy. Finally, the present corpus covers coding and writing as two common task ecologies. Other domains, populations, interfaces and deployment contexts may organize different opportunities for engagement and support.

These boundaries point to the next stage of research. The present study makes the interactional structure of AI-supported informal learning observable at scale; future work can test how these patterns translate into learning outcomes and design policies. Controlled interventions can examine how specific support forms influence users' constructive participation and downstream learning. Longitudinal user-level studies can ask whether repeated local episodes of feedback, explanation, revision and reflection become durable knowledge or transferable skill. Deployment studies can further examine how assistants should adapt support to user state, task ecology and interaction history in real-world settings. Within this scope, the study reframes everyday human–LLM interaction as a middle space between ordinary tool use and formal instruction. Together with formal instruction, personal AI assistants could support a more continuous and personalized learning ecosystem in which curricula provide structure, standards and shared progression, while assistants extend learning into everyday practice and help learners transform situated problem solving into durable understanding, transferable expertise and lifelong learning agency.

4 Methods

4.1 Study design

This study is an observational analysis of naturalistic public human–LLM conversations from WildChat, LMSYS Chat-1M and ShareChat strict-English across two everyday task ecologies, coding and writing. It asks whether learning-oriented engagement is visible in ordinary use, where constructive engagement is concentrated and how assistant scaffolding is ordered with user engagement across conversation-level and adjacent-turn timescales. The design is descriptive and associational, following the logic of large-scale digital trace and computational social-science analyses [26, 28]. It does not estimate randomized exposure effects, causal instructional efficacy or durable learning outcomes [27].

The empirical sequence mirrors the Results section. We first estimate the prevalence and composition of user engagement, then examine how constructive engagement varies by user framing, task ecology and interaction depth. We next test whether scaffolded assistant support marks conversations with richer constructive participation, separate scaffolded support by support form and analyse local temporal coupling between assistant support and user engagement. Dataset and task ecology are treated as substantive dimensions of the interaction environment rather than nuisance variables to be averaged away.

4.2 Data sources and analytic samples

The analytic corpus contains 128,334 conversations and 979,974 total turns, including 490,932 user turns and 489,042 assistant turns. The six task settings are WildChat coding, WildChat writing, LMSYS coding, LMSYS writing, ShareChat coding and ShareChat writing. WildChat and LMSYS Chat-1M provide large public human–LLM interaction logs from deployed chat systems, whereas ShareChat strict-English provides a smaller public-share corpus; together they offer complementary naturalistic settings rather than experimentally matched platform conditions [20–22]. WildChat coding contributes 31,878 conversations with 124,073 user turns and 124,623 assistant turns. WildChat writing contributes 39,534 conversations with 163,705 user turns and 164,824 assistant turns. LMSYS coding contributes 31,879 conversations with 106,312 user turns and 104,431 assistant turns. LMSYS writing contributes 21,023 conversations with 77,906 user turns and 76,279 assistant turns. ShareChat strict-English contributes 2,481 coding conversations with 11,541 user turns and 11,522 assistant turns and 1,539 writing conversations with 7,395 user turns and 7,363 assistant turns.

We also processed two newer corpora with the same pipeline as boundary checks. SWE-chat provides a recent agentic software-engineering setting and retained 1,536 coding conversations after filtering, so we report it as a coding-specific external check rather than as part of the balanced six-setting analysis [29]. ThoughtTrace retained only 56 coding and 61 writing conversations after the all-LLM v3 filter, with near-saturated scaffolded support, and is therefore kept as a feasibility check in the Appendix [30].

The six task settings were analysed with a common Level 1–4 annotation and analysis framework so that prevalence, contextual variation, support–engagement association and temporal ordering were computed from the same construct definitions. The manuscript reports the refreshed production outputs after the latest user-turn update pass and the subsequent LMSYS and ShareChat annotation runs rather than earlier pre-update batches. Supplementary Section A.1 describes the corpus construction and refresh provenance in more detail.

4.3 Conversation metadata and task ecologies

Each conversation is represented as an ordered sequence of user and assistant turns together with available metadata. The main contextual variables are data source, task domain, learning intent, topic and language. Coding and writing were selected because they are common everyday LLM use cases but differ in their interactional affordances: coding often involves debugging, error diagnosis and iterative testing, whereas writing often involves drafting, revision and directive editing [23–25]. Learning intent distinguishes conversations explicitly oriented toward understanding, exploration or learning from conversations in which learning-oriented engagement may be incidental to task completion.

Behavioural depth is measured from the number and ordering of turns in the conversation. Depth is used descriptively, as a marker of interactional opportunity and sustained exchange, not as an independent manipulation. Post-answer depth is measured as the number of turns after the first assistant response.

4.4 Turn-level constructs

The unit of annotation is the conversational turn. User and assistant turns are labelled with different constructs because they play different roles in the interaction. User turns are labelled for behavioural, emotional and cognitive engagement, drawing on engagement theory and the ICAP distinction between passive, active and constructive participation [14, 15]. The main analyses emphasize cognitive engagement and its depth levels. Passive engagement (`P`) captures reception or acknowledgement without substantial transformation. Active engagement (`A`) captures task-relevant action, follow-up or manipulation of supplied information. Constructive engagement (`C`) captures elaboration, revision, testing, extension or transformation of ideas. Constructive engagement is treated as the strictest behavioural proxy for learning-oriented participation available in these logs, not as a direct measure of knowledge gain.

Assistant turns are first distinguished as `S1` or `S2`. `S1` denotes a non-scaffolded reference response, including straightforward task fulfilment or stand-alone information provision. `S2` denotes scaffolded support: assistant behaviour that structures, regulates or redirects the user's process while leaving some cognitive work with the user. This definition follows the scaffolding tradition, especially contingency, transfer of responsibility and potential fading [16, 17, 19]. Scaffolded turns can also receive intention labels (`I1` metacognitive, `I2` cognitive and `I3` affective support) and non-mutually exclusive means labels (`M1` feedback, `M2` hinting, `M3` instructing, `M4` explaining, `M5` modelling and `M6` questioning). Operational examples are provided in Supplementary Table C1.

4.5 Annotation workflow and verification

The annotation workflow combined theory-grounded codebook development, prompt iteration, parser checks, human review and production-scale LLM-assisted labelling. We used LLM-assisted annotation as a scalable text-analysis workflow while retaining parser checks and human verification because prior work shows both the promise and the need for validation when language models are used as annotators [31, 32]. Prompt and post-processing rules were first developed on validation batches to reduce common failure modes, including over-labelling brief acknowledgements as active engagement and misclassifying directive assistant responses as scaffolding. After the pipeline stabilized, the same annotation logic was applied to the WildChat, LMSYS and ShareChat analytic corpora. All examples reproduced in the manuscript are de-identified and paraphrased rather than copied from raw logs.

Human verification was used as a quality-control input rather than as a separate source of outcome measurement. The final coding validation rerun combines three review rounds, with 658 task turns and 643 turns after deduplication. The final writing validation used 210 reviewed conversations. Agreement was summarized with per-label F1, Matthews correlation coefficient and Gwet's AC1 because the labels are sparse, prevalence-imbalanced and not mutually exclusive in all cases [33, 34]. Agreement was strongest for constructive, passive, emotional and support-form labels; boundary-heavy labels such as writing `A` and `S1` were less stable. Supplementary Table C2 reports the per-label agreement metrics, and Supplementary Section A.4 describes the final user-turn update check.

4.6 Outcome construction

The primary user-side outcome is constructive engagement. At the turn level, this is a binary indicator for whether the user turn is labelled `C`. At the conversation level, we use both the count of constructive user turns and an indicator for whether a conversation contains at least one constructive user turn. Constructive ratios are defined as constructive user turns divided by user turns.

The primary assistant-side exposure is scaffolded support. At the turn level, this is the distinction between `S2` scaffolded support and `S1` non-scaffolded reference response. At the conversation level, we use an indicator for whether a conversation contains at least one `S2` assistant turn. Support-form analyses condition on scaffolded turns and compare conversations containing each means label with scaffolded conversations that do not contain that label. Because means labels can co-occur, these are descriptive support-form contrasts rather than mutually exclusive treatment comparisons.

4.7 Statistical analyses

Descriptive analyses estimate engagement prevalence, scaffolding prevalence, passive/active/constructive composition, emotional-engagement rates and behavioural-depth summaries within each public-chat task setting. Context analyses compare intentional versus unintentional conversations and coding versus writing settings. Depth analyses group conversations into behavioural-depth buckets and estimate the probability that a conversation contains at least one constructive turn.

Conversation-level associations are analysed by comparing conversations with and without at least one scaffolded assistant turn. Constructive-turn counts are modelled with Poisson generalized linear models. Binary outcomes, including whether a conversation contains any constructive turn, whether the next user turn is constructive and whether the next assistant turn is scaffolded, are modelled with logistic regression or reported as conditional probability contrasts, depending on the estimand [35, 36]. Adjusted models include observed covariates appropriate to the analysis level, including learning intent, task domain or topic indicators, turn counts, emotional ratios, error-related variables and turn index. Percentage-point contrasts are reported as effect sizes; their uncertainty is estimated with conversation-level bootstrap procedures, using conversation-cluster bootstrap resampling for adjacent-turn analyses because multiple turn pairs can come from the same conversation. Supplementary Tables C3–C5 summarize the main model families and temporal estimands.

To assess whether the adjacent-turn pattern was reducible to a single feature family, we also fitted integrated logistic regressions that combined user-state features, assistant-scaffolding features and adjacent-turn context in one model. The outcome was whether the next user turn was constructive. The pooled model included dataset fixed effects; a sensitivity replaced these with recoverable assistant model or source-family fixed effects, using exact `chat_model` labels where available, LMSYS model names recovered from conversation identifiers and ShareChat assistant/source families recovered from public-share identifiers. Standard errors for these adjacent-turn regressions were clustered by conversation. These model/source checks are interpreted as robustness analyses rather than model-causal comparisons.

Temporal analyses use two complementary scales. Adjacent-turn analyses compare the probability that the next user turn is constructive after scaffolded versus non-scaffolded assistant turns and the probability that the next assistant turn is scaffolded after different prior user states. Coarser within-conversation analyses compare constructive engagement before versus after the first scaffolded assistant turn and first-half versus second-half summaries where available. The adjacent-turn analyses test local ordering; the coarser analyses test whether there is evidence for longer-span accumulation.

4.8 Robustness, privacy and LLM use

WildChat robustness analyses repeat the main patterns across user-facing model snapshots and coarse model families. These checks assess whether the headline patterns are reducible to a single WildChat model label. They are not interpreted as model-causal comparisons, because model labels are not experimentally assigned and may be confounded with time, user population, task mix and logging context.

The analyses are bounded by the observational design and by the nature of the labels. They support claims that everyday human–LLM conversations contain measurable learning-oriented engagement; that constructive engagement varies by intent, task ecology and depth; that assistant scaffolding is associated with richer constructive participation; and that support and engagement show local reciprocal ordering. They do not support claims that scaffolded support causes durable learning gains, that all constructive engagement reflects actual knowledge change or that the labelled conversational states perfectly measure learning itself.

LLMs were used as objects of study and as annotation tools. Production labels were generated with LLM-assisted workflows and then checked through parser rules, targeted update scripts and human validation. The authors also used LLM-based tools to assist with code drafting, debugging, analysis workflow organization and language editing. LLM-generated assistance was not used as a substitute for human verification of the validation samples or for interpretation of the statistical results.

All reported results are aggregate turn-level or conversation-level analyses. The study does not link conversations into user histories, estimate user-level longitudinal trajectories or publish raw conversation text; temporal analyses are confined to ordering within a conversation.

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Supplementary information

Supplementary information is provided below.

Appendix A Supporting methods

A.1 Corpus construction and analytic settings

The analyses use six task settings formed by crossing three public conversational corpora with two task ecologies: WildChat coding, WildChat writing, LMSYS coding, LMSYS writing, ShareChat coding and ShareChat writing. Coding and writing were selected because both are common everyday LLM use cases, yet they differ in the opportunities they create for visible learning-oriented engagement. Coding conversations often contain error messages, debugging attempts, execution failures and iterative repair. Writing conversations more often involve drafting, rewriting, shortening, tone adjustment and direct editing.

The analytic corpus contains 128,334 conversations and 979,974 total turns. WildChat contributes 31,878 coding conversations and 39,534 writing conversations; LMSYS contributes 31,879 coding conversations and 21,023 writing conversations; ShareChat strict-English contributes 2,481 coding conversations and 1,539 writing conversations. All reported analyses use the refreshed Level 1–4 outputs after the latest user-turn update pass and the subsequent LMSYS and ShareChat annotation runs.

SWE-chat and ThoughtTrace were also processed with the same pipeline as feasibility and boundary checks (Supplementary Table C9) [29, 30]. SWE-chat retained a usable coding sample (1,536 conversations) but only three writing conversations after filtering, so it can serve only as a coding-specific stress test rather than as part of the main coding–writing comparison. ThoughtTrace retained 56 coding and 61 writing conversations after the all-LLM v3 filter, but the sample remains small and scaffolded support is near-saturated, especially in coding where all conversations contain at least one scaffolded assistant turn. These properties make ThoughtTrace useful as a feasibility check but not as stable evidence for conversation-level support-association estimation.

For each conversation, the pipeline retained the ordered turn sequence, role labels, available topic and language metadata, learning-intent indicators and model metadata where available. Model metadata are used only for robustness checks where the corpus exposes usable model labels, because model assignment in naturalistic public conversations is observational and may reflect time, user population, task mix and logging context.

A.2 Annotation ontology

The codebook translates learning-science constructs into role-specific behavioural signatures. User turns are labelled for engagement. Passive engagement (`P`) marks acknowledgement, reception or minimal continuation without visible transformation. Active engagement (`A`) marks task-relevant action or follow-up, such as asking for a reformatted answer or applying supplied information. Constructive engagement (`C`) marks deeper work with ideas, such as testing an implication, revising a hypothesis, articulating a constraint, transferring a mechanism or extending an explanation. Emotional engagement is labelled separately because affective expressions were rare and not the main analytic outcome.

Assistant turns are labelled for support. `S1` denotes a non-scaffolded reference response: the assistant provides the requested information or deliverable without materially structuring the user's process. `S2` denotes scaffolded support: the assistant structures, regulates, diagnoses or redirects the user's work while leaving some responsibility with the user. Scaffolded turns can also receive support-intent labels and support-means labels. Intention labels distinguish metacognitive support (`I1`), cognitive support (`I2`) and affective support (`I3`). Means labels distinguish feedback (`M1`), hinting (`M2`), instructing (`M3`), explaining (`M4`), modelling (`M5`) and questioning (`M6`). Means labels are non-mutually exclusive because a single assistant turn can both explain a mechanism and provide an example or hint. Supplementary Table C8 reports the support-means supply profile across the six task settings.

A.3 LLM-assisted annotation pipeline

The annotation pipeline combined prompt-based labelling with deterministic parser checks and targeted post-processing. During development, prompts were iterated against validation examples and disagreement cases. Common failure modes included treating brief acknowledgements as active engagement, treating any long assistant answer as scaffolding, collapsing directive instructions into scaffolding without checking whether responsibility remained with the user, and over-labelling sparse support means.

Production labels were generated at turn level. Parser checks enforced the expected label schema, normalized label variants and flagged missing or malformed outputs. Post-processing rules were used only when a recurring error pattern could be specified transparently, such as a narrow class of user-turn false positives. Prompt versions, run metadata, disagreement files and update outputs were retained with the analysis artifacts so that validation cases can be traced without reproducing raw conversation text in the manuscript.

A.4 Human verification and update provenance

Human verification was used to quantify label reliability and identify boundary cases. Coding validation combined three review rounds, with 658 task turns and 643 turns after deduplication. Writing validation used 210 reviewed conversations. Because many labels are sparse or prevalence-imbalanced, agreement is reported with per-label F1, Matthews correlation coefficient and Gwet's AC1 rather than with a single global agreement statistic.

The final user-turn check contained 100 coding examples and 100 writing examples reviewed after the update pass. These examples were used to identify residual false positives and calibrate patch-style update filters, especially for Active and Passive user-turn labels. The last WildChat user-turn update pass scanned 31,878 coding conversations and 39,534 writing conversations, applying 1,815 coding updates and 21 writing updates. LMSYS and ShareChat were then processed with the stabilized replacement-data workflow using the same construct definitions and post-processing logic. The main manuscript reports the refreshed labels produced by these final production runs.

A.5 Outcome construction

At the turn level, the focal user-side outcome is a binary indicator for constructive engagement. At the conversation level, we use both the count of constructive user turns and an indicator for whether at least one constructive user turn occurs. Constructive ratios divide constructive user turns by total user turns. Cognitive engagement ratios include passive, active and constructive cognitive engagement.

At the assistant side, scaffolded support is represented by the `S2` label. Conversation-level scaffolding presence indicates whether at least one assistant turn in the conversation is labelled `S2`. Support-form analyses condition on conversations or turns with scaffolded support and then compare the presence versus absence of each means label. These comparisons are descriptive because means labels can co-occur and because conversations are not randomly assigned to receive a support form.

Behavioural depth is computed from turn counts and turn ordering. Post-answer depth counts turns after the first assistant response. Persistence after failure is analysed only for coding-oriented conversations because debugging failures, error messages and failed-execution contexts are more directly observable in coding than in writing.

A.6 Statistical model specification

The main analyses use simple estimands matched to the claim being made. Descriptive prevalence analyses report proportions within the six public-chat task settings. Context analyses compare learning intent, task ecology and depth groups. Conversation-level support associations use Poisson generalized linear models for constructive-turn counts and logistic models for binary constructive-engagement outcomes. Adjacent-turn analyses use assistant-to-user and user-to-assistant turn pairs to estimate local conditional probabilities.

Covariates are included when they correspond to observed compositional differences at the analysis level. Depending on the analysis, these include learning intent, topic or task indicators, turn counts, emotional ratios, error-related variables and turn index. Percentage-point contrasts are reported as effect sizes. Confidence intervals and p values for these contrasts use conversation-level bootstrap resampling; adjacent-turn contrasts resample conversations rather than individual turn pairs to account for repeated pairs within conversations.

The integrated adjacent-turn regression combines three feature groups in one logistic model: user state before the assistant turn, assistant scaffolding features, and adjacent-turn context. The pooled specification includes dataset fixed effects. A model/source sensitivity uses exact WildChat model labels, LMSYS model names recovered from conversation identifiers and ShareChat assistant/source families recovered from public-share identifiers. These adjusted estimates are interpreted as measured-covariate-adjusted associations. They do not remove unobserved user motivation, prior expertise, task difficulty or time-varying conversational state.

A.7 Temporal scale

The temporal analyses deliberately separate adjacent-turn ordering from coarser within-conversation summaries. The adjacent-turn analyses ask whether the immediately following user turn is more likely to be constructive after scaffolded support than after a non-scaffolded reference response, and whether that contrast depends on the prior user state. The reverse adjacent-turn analysis asks whether the assistant is more likely to provide scaffolded support after constructive, active or passive user turns.

The coarser analyses ask a different question: whether constructive engagement is higher after the first scaffolded assistant turn than before it, and whether constructive engagement increases from the first to the second half of a conversation. These are not treatment-effect estimates. The first scaffolded turn may occur precisely because the user is struggling, because a difficult task has emerged or because the exchange is already developing. The main temporal claim is therefore scale-specific: the clearest evidence concerns local, state-conditioned coupling rather than cumulative uplift.

Appendix B Robustness and interpretive boundaries

B.1 WildChat model-snapshot robustness

WildChat includes user-facing model metadata that make it possible to repeat major descriptive patterns across model snapshots and coarse model families. These analyses preserved the main directional patterns reported in the main text: coding generally showed higher cognitive engagement than writing; intentionality gaps were visible across most user-facing models; scaffolded conversations were generally more constructive; and adjacent-turn contrasts were more interpretable than coarser within-conversation contrasts. Supplementary Figures D3–D7 report these WildChat model-snapshot checks.

These checks are used as robustness analyses rather than model-causal comparisons. WildChat model labels are not experimentally assigned. They may be confounded with calendar time, user population, task mix, interface changes and logging context. For this reason, the manuscript does not claim that a specific model family caused the observed engagement differences.

B.2 Label uncertainty

The label scheme intentionally separates top-level constructs from finer support forms. Top-level labels such as constructive engagement and scaffolded support have clearer behavioural anchors. Fine-grained labels, especially sparse or boundary-heavy labels, are less stable. This is why the main claims rely most heavily on constructive engagement, scaffolded support presence, support-form patterns that are consistent across settings and temporal contrasts that can be expressed with simple conditional probabilities.

B.3 Selection into support

The most important threat to causal interpretation is selection into support. More complex tasks, more motivated users or already-developing exchanges may both elicit scaffolded support and produce constructive follow-up. Conversation-level adjusted models reduce measured compositional differences but cannot remove unobserved selection. Adjacent-turn analyses reduce the temporal window and condition on local state, but they still cannot identify randomized exposure effects. The safest interpretation is therefore local, state-dependent coupling between assistant support and user engagement.

B.4 Privacy-preserving reporting

The study analyses naturalistic conversations, so the manuscript avoids reproducing raw conversation text. Operational examples are de-identified and paraphrased to illustrate decision rules without exposing user content. Aggregate statistics, model summaries and validation metrics are reported in the manuscript and supplementary tables. The analyses do not link conversations into user histories or estimate longitudinal user-level trajectories; temporal claims are restricted to within-conversation turn ordering.

Appendix C Supporting tables

Table C1 Operational examples for turn-level labels. Examples are de-identified and paraphrased from validation review patterns; they illustrate the decision rule rather than reproducing raw conversation text.

Construct	Paraphrased turn	Annotation rationale
`P` passive user engagement	"Thanks, that makes sense."	Minimal uptake or acknowledgement without new reasoning, evaluation or task transformation.
`A` active user engagement	"Can you convert that answer into a command I can run in my setup?"	Task-relevant follow-up or manipulation of supplied information, but without elaborating or testing an idea.
`C` constructive user engagement	"If the cache expires before the asynchronous callback returns, would the same race condition still happen?"	The user tests an implication, boundary condition or transfer of the prior explanation.
`S1` non-scaffolded reference	"Here is the requested function/email draft."	The assistant primarily completes the requested deliverable without regulating the user's process.
`S2` scaffolded support	"First isolate the failing step; that will show whether the issue is parsing or state management."	The assistant structures the user's process and leaves diagnostic or revision work with the user.
`M1` feedback	"Your diagnosis is close; the bug is in the index offset, not the loop boundary."	Evaluative feedback on the user's prior attempt or understanding.
`M2` hinting	"Check the parser log around the first malformed token before changing the model."	A cue or partial path that helps the user proceed without fully resolving the task.
`M3` instructing	"First split the condition into two branches, then add a regression test."	Explicit steps or strategy that organizes the user's work.
`M4` explaining	"The exception occurs because the event loop is already running, so a nested call cannot acquire it."	A rationale or mechanism that clarifies why the issue occurs.
`M5` modeling	"A small pattern you can adapt is: validate input, normalize it, then pass it to the writer."	A sample approach intended to be emulated or adapted rather than merely copied.
`M6` questioning	"Which constraint matters more here: preserving tone or shortening the paragraph?"	A question that advances problem solving or reflection.

Table C2 Human–human agreement for labels used in the main analyses. Coding uses the latest three-round validation rerun over 658 task turns, 643 after deduplication; writing uses the latest writing-210 best-pipeline validation. F1, Matthews correlation coefficient (MCC) and Gwet's AC1 are reported per binary label. Coding MCC and AC1 are recomputed from the exported support and positive-count fields; writing metrics are from the exported agreement report. Rare degenerate labels with no positive cases in a validation set are omitted.

Domain	Role	Label	F1	MCC	Gwet AC1	Support
Coding	User	Active (`A`)	0.75	0.72	0.87	315
Coding	User	Constructive (`C`)	0.85	0.82	0.90	315
Coding	User	Passive (`P`)	0.88	0.87	0.96	315
Coding	User	Emotional (`E`)	0.83	0.82	0.97	315
Coding	Assistant	Non-scaffolded reference (`S1`)	0.78	0.73	0.88	328
Coding	Assistant	Scaffolded support (`S2`)	0.87	0.69	0.70	328
Coding	Assistant	Feedback (`M1`)	0.90	0.89	0.95	328
Coding	Assistant	Hinting (`M2`)	0.79	0.75	0.89	328
Coding	Assistant	Instructing (`M3`)	0.81	0.77	0.92	328
Coding	Assistant	Explaining (`M4`)	0.86	0.79	0.84	328
Coding	Assistant	Modeling (`M5`)	0.84	0.79	0.87	328
Coding	Assistant	Questioning (`M6`)	0.82	0.81	0.95	328
Writing	User	Active (`A`)	0.38	0.38	0.69	90
Writing	User	Constructive (`C`)	0.89	0.83	0.86	90
Writing	User	Passive (`P`)	0.80	0.81	0.99	90
Writing	User	Emotional (`E`)	0.67	0.65	0.96	90
Writing	Assistant	Non-scaffolded reference (`S1`)	0.46	0.43	0.86	120
Writing	Assistant	Scaffolded support (`S2`)	0.97	0.80	0.93	120
Writing	Assistant	Feedback (`M1`)	0.89	0.80	0.80	120
Writing	Assistant	Hinting (`M2`)	0.82	0.70	0.71	120
Writing	Assistant	Instructing (`M3`)	0.71	0.67	0.86	120
Writing	Assistant	Explaining (`M4`)	0.84	0.70	0.70	120
Writing	Assistant	Modeling (`M5`)	0.94	0.91	0.95	120
Writing	Assistant	Questioning (`M6`)	0.96	0.96	0.99	120

Table C3 Analytic model families used in the main analyses. The table summarizes descriptive and conversation-level model families. Adjusted models are interpreted as covariate-adjusted associations in naturalistic data, not as causal treatment-effect estimates.

Analysis family	Unit and outcome	Main contrast or model	Scope
Engagement prevalence	User turns and conversations; cognitive, constructive and emotional engagement indicators.	Descriptive proportions by public-chat task setting, task ecology and learning intent.	Establishes where learning-oriented engagement is visible; no causal contrast is implied.
Engagement composition	Cognitively engaged user turns; passive, active and constructive depth levels.	Composition of cognitive engagement within each public-chat task setting.	Motivates constructive engagement as the focal strict outcome while preserving the broader cognitive-engagement hierarchy.
Interaction depth	Conversations; at least one constructive user turn.	Probability of constructive engagement across behavioural-depth buckets.	Depth is interpreted as interactional opportunity and task development, not as a manipulated exposure.
Scaffolding presence	Conversations; constructive-turn count and any constructive turn.	Poisson generalized linear models for counts and logistic models for binary outcomes, comparing conversations with versus without scaffolded assistant turns.	Adjusted estimates reduce measured compositional differences but remain associational because support is selected into by task difficulty and user state.

Table C4 Additional analytic model families and robustness checks. These analyses qualify the main associations by support form, temporal scale and available model or source metadata.

Analysis family	Unit and outcome	Main contrast or model	Scope
Support-form associations	Scaffolded conversations; constructive engagement ratios.	Mean differences for conversations containing a given support-means label versus scaffolded conversations without that label, summarized by user framing and task ecology.	Descriptive because means labels are non-exclusive and can co-occur in the same assistant turn.
Local temporal coupling	Adjacent assistant–user or user–assistant turn pairs; next constructive user turn or next scaffolded assistant turn.	Conditional probability contrasts around adjacent turns; detailed estimands are listed in Supplementary Table C5.	Interpreted as local ordering and state-dependent coupling, not as randomized support effects.
Integrated adjacent-turn regression	Adjacent assistant–user turn pairs; next constructive user turn.	Logistic regression combining prior user state, scaffolded-support presence, support means, task/framing context and dataset or model/source fixed effects.	Tests whether the local coupling pattern is reducible to one feature family; standard errors are clustered by conversation.
Coarse temporal contrasts	Conversations containing scaffolded support; constructive engagement before and after the first scaffolded turn.	Mean after-minus-before contrast and first-half versus second-half robustness summaries where available.	Tests longer-span accumulation patterns, but the first scaffolded turn may be selected by difficulty, uncertainty or an already developing exchange.
WildChat model robustness	WildChat conversations or turns; main prevalence, association and temporal outcomes.	Repetition across user-facing model snapshots or coarse model families.	Robustness summaries only; model labels are observational and may be confounded with time, user mix and task mix.

Table C5 Temporal estimands used to separate local coupling from coarse accumulation. The analyses intentionally combine adjacent-turn and longer-span summaries because the theoretical claim concerns temporal scale. Adjacent-turn contrasts test local ordering; within-conversation contrasts test whether there is evidence for broader accumulation after scaffolding appears.

Estimand	Definition	Reported in main text	Scope
Overall adjacent-turn lift	Difference in the probability that the next user turn is constructive after scaffolded support versus after a non-scaffolded reference response, estimated within each public-chat task setting.	Figure 5a	Local association between assistant support type and immediately following constructive engagement; not a randomized support effect.
Prior-state adjacent contrast	Difference in next-turn constructive probability after scaffolded support versus reference response, separately for prior constructive, active and passive user states.	Figure 5b	Tests whether local support–engagement coupling depends on the user's immediately preceding state.
Reverse state-conditioned support	Probability that the next assistant turn is scaffolded, estimated after constructive, active and passive user turns.	Figure 5c	Describes how assistant scaffolding supply varies with the user's current engagement state.
Around-first-scaffold contrast	Mean constructive engagement after the first scaffolded assistant turn minus mean constructive engagement before that turn, among conversations containing scaffolded support.	Figure 5d	Coarse before–after contrast; the first scaffolded turn may be selected by difficulty, uncertainty or an already developing exchange.
First-half versus second-half contrast	Constructive engagement in the second half of a conversation minus constructive engagement in the first half, reported where available in the production outputs.	Results text	A longer-span robustness lens; increases do not identify scaffolding as the cause of later engagement.

Table C6 Uncertainty estimates for key percentage-point contrasts. Constructive-ratio contrasts are conversation-bootstrap estimates weighted by user turns; adjacent-turn contrasts bootstrap conversations over assistant-to-user pairs. Percentage-point differences are the effect sizes reported in the main figures.

Setting	Contrast	Effect	95% CI	p value
WC coding	Has S2 − no S2 constructive ratio	+4.56 pp	[+4.15, +4.99]	< .001
LMSYS coding	Has S2 − no S2 constructive ratio	+2.51 pp	[+2.19, +2.86]	< .001
SC coding	Has S2 − no S2 constructive ratio	+7.34 pp	[+5.29, +9.27]	< .001
WC writing	Has S2 − no S2 constructive ratio	+0.99 pp	[+0.82, +1.16]	< .001
LMSYS writing	Has S2 − no S2 constructive ratio	+1.09 pp	[+0.87, +1.31]	< .001
SC writing	Has S2 − no S2 constructive ratio	+3.20 pp	[+1.55, +4.89]	< .001
WC coding	Adjacent S2 − reference lift	+2.68 pp	[+2.22, +3.09]	< .001
LMSYS coding	Adjacent S2 − reference lift	+2.01 pp	[+1.54, +2.55]	< .001
SC coding	Adjacent S2 − reference lift	+6.21 pp	[+4.35, +8.10]	< .001
WC writing	Adjacent S2 − reference lift	+1.13 pp	[+0.88, +1.41]	< .001
LMSYS writing	Adjacent S2 − reference lift	+0.27 pp	[-0.01, +0.56]	.061
SC writing	Adjacent S2 − reference lift	+3.35 pp	[+0.80, +6.62]	.025

Table C7 Integrated adjacent-turn regression combining user state, assistant scaffolding and model controls. The outcome is whether the next user turn is constructive. Estimates are odds ratios with 95% confidence intervals and two-sided p values from conversation-cluster robust standard errors. The primary pooled model uses dataset fixed effects; the model/source sensitivity adds recoverable assistant model or source-family fixed effects.

Predictor	Pooled dataset FE	Pooled model/source FE	WildChat model FE
Scaffolded support	0.94 [0.87, 1.02], .164	0.93 [0.85, 1.00], .063	0.98 [0.89, 1.08], .689
Prior user constructive	8.24 [7.81, 8.69], < .001	7.94 [7.53, 8.38], < .001	8.48 [7.92, 9.08], < .001
Prior user active	2.47 [2.37, 2.57], < .001	2.43 [2.33, 2.53], < .001	2.69 [2.56, 2.84], < .001
Prior user passive	2.15 [1.83, 2.53], < .001	2.14 [1.82, 2.52], < .001	2.36 [1.89, 2.96], < .001
Intentional framing	1.37 [1.32, 1.43], < .001	1.39 [1.34, 1.45], < .001	1.37 [1.30, 1.44], < .001
Coding task	1.34 [1.19, 1.50], < .001	1.34 [1.20, 1.51], < .001	1.16 [1.01, 1.34], .039
M1 feedback	0.79 [0.73, 0.86], < .001	0.80 [0.74, 0.86], < .001	0.74 [0.67, 0.82], < .001
M4 explaining	2.15 [1.99, 2.32], < .001	2.12 [1.96, 2.30], < .001	2.11 [1.92, 2.31], < .001
M6 questioning	0.82 [0.72, 0.93], .002	0.84 [0.74, 0.96], .008	1.08 [0.93, 1.25], .340
Model/source block	Reference	LR $\chi^2_{41} = 718.2$, $p < .001$	LR $\chi^2_{13} = 553.2$, $p < .001$

Table C8 Support-supply profile within scaffolded assistant turns. Values show the percentage of scaffolded assistant turns containing each scaffolding-means label. Means labels are not mutually exclusive, so rows do not sum to 100%. Coding-oriented scaffolded turns are dominated by explanation-heavy support, whereas writing-oriented scaffolded turns show a more mixed support profile.

Setting	M1 Feedback	M2 Hinting	M3 Instructing	M4 Explaining	M5 Modelling	M6 Questioning
WildChat coding	8.8	14.7	29.7	84.0	23.9	6.9
LMSYS coding	4.5	8.2	10.4	79.2	18.3	10.6
ShareChat coding	13.3	12.7	25.7	81.2	18.4	18.6
WildChat writing	16.6	9.6	59.2	29.5	15.0	6.9
LMSYS writing	11.3	10.9	52.3	26.9	10.3	13.4
ShareChat writing	24.2	23.2	22.9	45.2	23.5	38.1

Table C9 Boundary-check corpora processed with the common pipeline. SWE-chat and ThoughtTrace were labelled and analysed with the same Level 1–4 workflow, but are not part of the main six-setting analysis because they do not provide stable coding–writing task comparisons or stable non-scaffolded reference variation. Percentages for cognitive, constructive and emotional engagement use user turns as the denominator; scaffolded-support percentages use assistant turns as the denominator.

Corpus / setting	Use in manuscript	Conv.	User turns	Assistant turns	Scaffolded turns (%)	Cognitive overall (%)	Constructive level (%)	Boundary-check implication
SWE-chat coding	Coding-only appendix check	1,536	9,506	6,813	19.45	32.36	17.96	Scaffolded conversations were more constructive than non-scaffolded conversations (0.173 vs. 0.133) and deeper after the first answer (+6.41 turns), but the corpus lacks a usable writing counterpart.
SWE-chat writing	Excluded from inference	3	18	14	42.86	22.22	11.11	Only three conversations remained after filtering, so estimates are not interpretable.
ThoughtTrace coding	Feasibility check only	56	251	252	85.71	64.54	9.96	All conversations contained scaffolded support, leaving no non-scaffolded conversation-level reference group for support-association estimates.
ThoughtTrace writing	Feasibility check only	61	301	300	67.33	51.16	2.66	Scaffolded support remained very common and only two conversations lacked scaffolded support, making adjusted support contrasts unstable.

Appendix D Supporting figures

The supporting figures report extended temporal checks and model-snapshot robustness analyses where model metadata are available. They are included to make the main claims auditable without shifting the manuscript's emphasis away from the six main public-chat task settings. Model-snapshot figures are descriptive robustness checks because model labels in naturalistic corpora are observational rather than experimentally assigned.

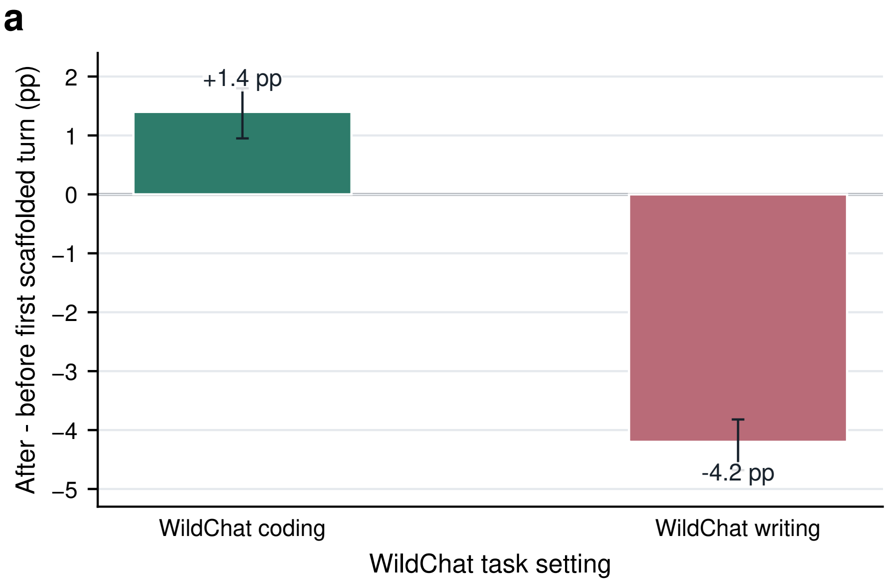

Fig. D1 Coarse around-first-scaffold contrasts provide limited evidence for cumulative improvement. Bars show the difference in constructive engagement after versus before the first scaffolded assistant turn within the WildChat robustness subset of conversations containing scaffolded support. The contrasts are mixed across the two WildChat task settings, supporting the main-text interpretation that the clearest temporal evidence concerns local adjacent-turn coupling rather than broad cumulative uplift.

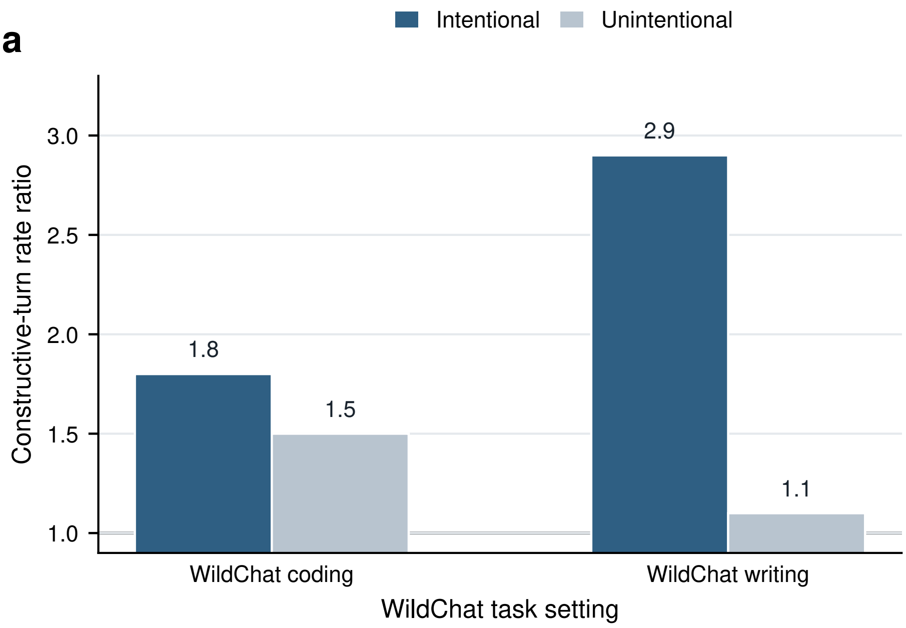

Fig. D2 Scaffolded-support associations vary by user framing. Stratified model estimates compare the association between scaffolded-support presence and constructive participation in intentional versus unintentional conversations. The figure is interpreted as descriptive moderation rather than evidence that intent causally changes support effectiveness, because user framing, task complexity and persistence are not randomly assigned.

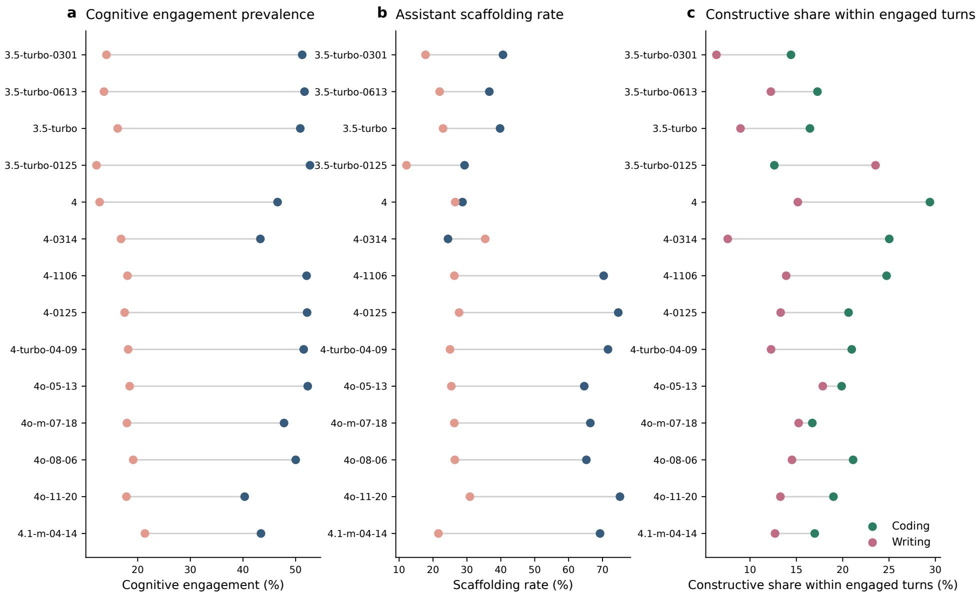

Fig. D3 WildChat model-snapshot prevalence profiles preserve the main domain contrast. Across user-facing WildChat model snapshots, coding-oriented conversations generally show higher cognitive engagement and higher assistant scaffolding rates than writing-oriented conversations. The model labels are observational and may reflect time, user mix and traffic composition, so the figure is used as a robustness check rather than a model-causal comparison.

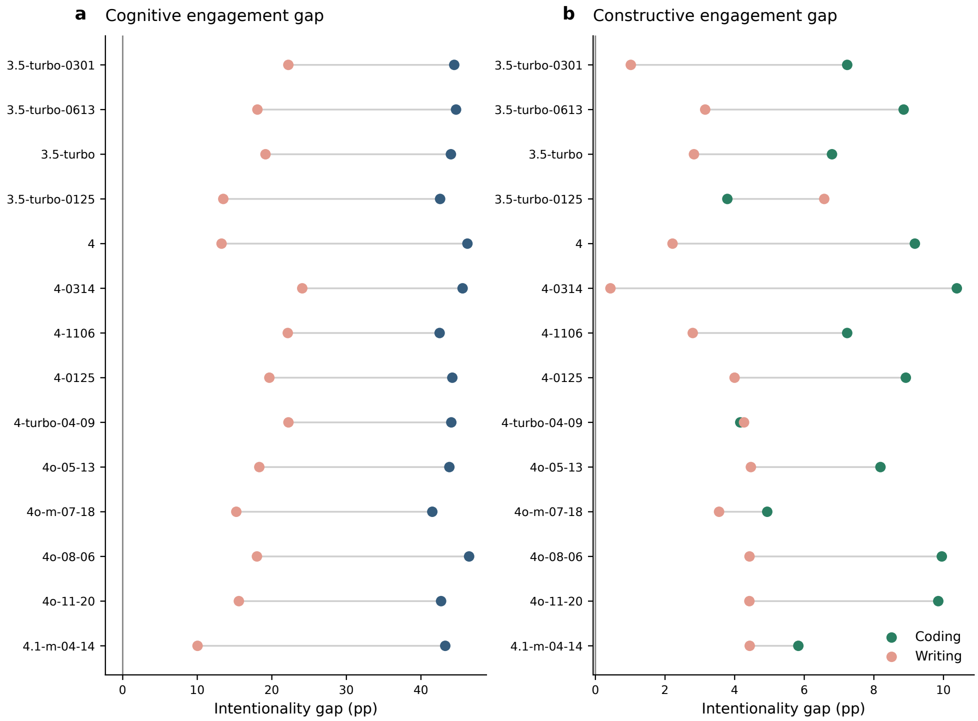

Fig. D4 Intentionality gaps are visible across WildChat model snapshots. The figure reports model-specific differences between intentional and unintentional conversations in cognitive and constructive engagement, separately for coding and writing. Most model snapshots preserve the directional pattern that intentionally framed conversations contain richer learning-oriented participation.

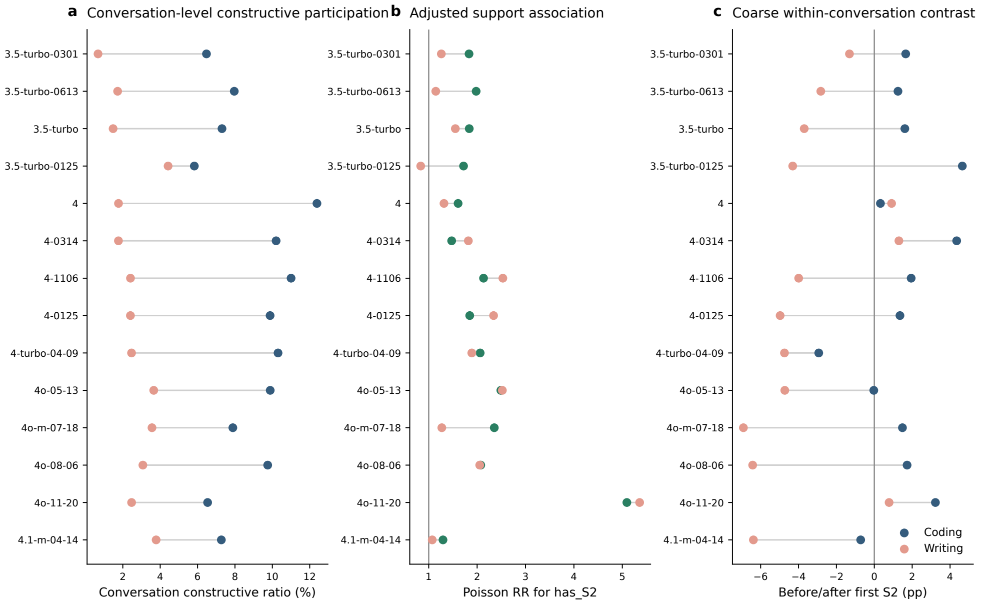

Fig. D5 Conversation-level support associations are broadly positive across WildChat model snapshots. The figure repeats major conversation-level analyses by user-facing WildChat model snapshot, including constructive participation, adjusted support associations and coarse before–after contrasts around the first scaffolded assistant turn. The pattern supports the distinction between stable conversation-level associations and more mixed coarse temporal contrasts.

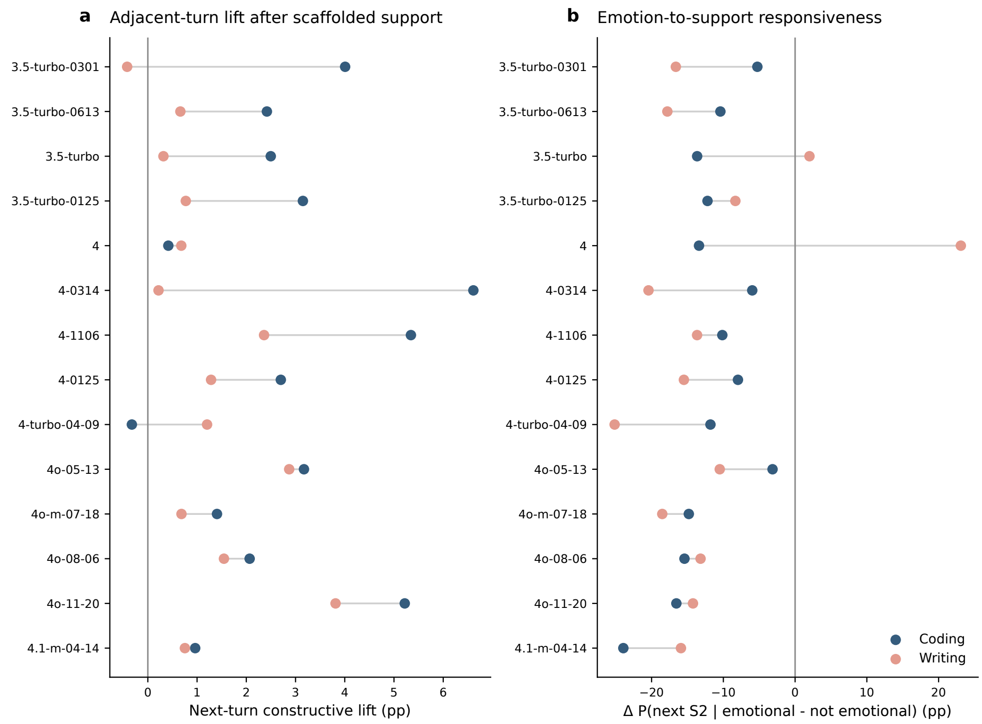

Fig. D6 Local temporal signals vary across WildChat model snapshots. The figure reports adjacent-turn constructive lifts and emotion-to-support responsiveness by model snapshot. Adjacent-turn constructive lifts are often positive, especially in coding-oriented conversations, but the magnitude varies across snapshots; emotion-to-support responsiveness is more heterogeneous and is treated as exploratory.

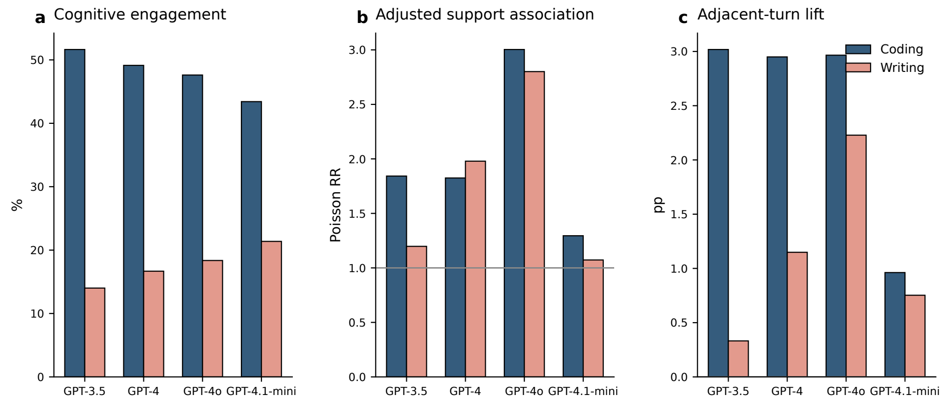

Fig. D7 Coarse model-family aggregation preserves the main directional patterns. WildChat model snapshots are grouped into coarse model families to summarize cognitive engagement prevalence, adjusted support associations and adjacent-turn constructive lift. Coding remains higher than writing across families, and the support-association and local-lift patterns remain directionally consistent on average.